Low Voltage Sensing and Indication

Hitesh Nandakumar
Computer Engineering Department
California Polytechnic State University, San Luis Obispo
San Luis Obispo, California, United States of America
hnandaku@calpoly.edu

***Abstract*— This paper presents the design, implementation, and experimental characterization of a low-supply-voltage sensor with an optical indicator. Using a Zener-based voltage limiter, a linear voltage divider, and a TLC3702 comparator, the circuit detects when the supply voltage drops below a designed threshold near 8 V and activates an LED warning indicator. Measurements confirm the expected Zener diode behavior, comparator switching threshold, and LED activation, demonstrating reliable low-voltage sensing.**

***Keywords*— Zener diode, comparator, LED indicator, reverse-bias, breakdown voltage, analog circuits, undervoltage detection**

I. Introduction

Low-voltage detection is critical in systems where supply rails must remain above minimum levels to ensure proper operation. In this experiment, a low voltage detector is constructed using a Zener diode reference and a TLC3702 comparator [1]. A red LED provides a visual indicator when the supply voltage drops below the designed threshold. The laboratory objective is to both implement a given low supply sensor-indicator circuit and compare theoretical expectations with measured circuit behavior.

A. Zener-based Voltage Reference

A Zener diode operated in reverse breakdown provides a nearly constant reference voltage V_Z [1]. When the diode current exceeds the minimum knee current, the breakdown voltage remains approximately constant. This property is used in the lab circuit to create a fixed comparison voltage for the TLC3702 comparator [1].

In this experiment, the Zener diode and its series resistor form a reference branch whose voltage remains approximately constant even as the supply varies. This stable reference is compared against the scaled supply voltage produced by the linear divider described in the lab manual [1], enabling reliable detection of the point at which the supply drops below the designed trip voltage.

The aforementioned series resistor's resistance is determined by the following equation:

$$R_Z < \frac{v_{DD}-v_Z}{I_{Zk}} \tag{1}$$

In this equation, the knee current I_{Zk} represents the minimum current required to maintain stable breakdown; in this design, a value of 0.25 mA was used, consistent with device specifications [3], ensuring proper reference operation under all measured supply conditions.

B. Resistive Divider and Comparator Threshold

The resistive divider sets the comparator's variable input by scaling the supply voltage through two resistors arranged in series. This divider produces a fraction of the supply voltage, given by the equation:

$$V_+ = \frac{R_a}{R_a+R_b} V_{DD} \tag{2}$$

The output of which decreases proportionally as the supply drops. As described in the lab handout [1], this node serves as the comparator's positive input and represents the circuit's adjustable sensing threshold.

The relationship between the divider ratio and the Zener reference determines the precise voltage at which the comparator switches. When the scaled divider output falls below the Zener reference voltage V_Z, the TLC3702 comparator transitions its output state [4]. The trip condition can be approximated by the equation:

$$V_{DD} = \left(1 + {}^{R_b}/_{R_a}\right) V_Z \tag{3}$$

The equation shows directly how the resistor ratio sets the desired threshold. This selection of and allows the divider to be tuned so that the comparator switches near 8 V, enabling low-supply detection.

II. Experimental Procedure

A. Procedure 1 – Zener Voltage Limiter

The first stage of the experiment involved constructing the Zener-based voltage limiter on a breadboard as outlined in the lab handout [1]. The circuit consisted of the 1N5229B Zener diode in series with a bias resistor. A Rigol DP832 Power Supply provided input voltages ranging from 7 V to 10 V in 0.5

V increments. At each step, the voltage across the Zener diode was measured and recorded.

TABLE I: Zener Diode Voltage from Procedure 1 Circuit

V_{DD}, V	**7**	**7.5**	**8.0**	**8.5**	**9.0**	**9.5**	**10**
V_Z, mV	3400	3440	3480	3520	3560	3600	3620

The resulting measurements showed the expected slight upward slope in Zener voltage, increasing from 3400 mV at 7.0 V supply to 3620 mV at 10.0 V.

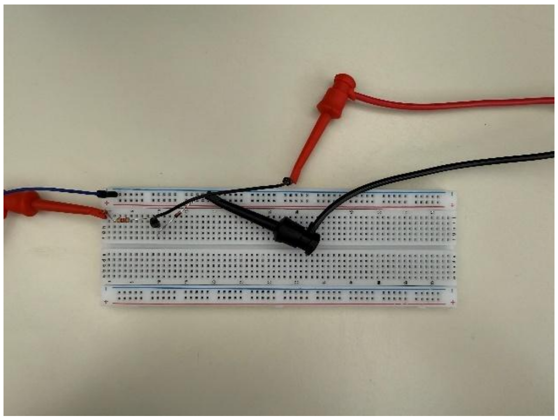

Fig. 1: Construction of Circuit from Procedure 1

A photograph of the assembled limiter circuit was taken to document construction quality and measurement setup.

B. *Procedure 2 – Divider Calibration*

Next, the adjustable resistive divider was added to the limiter circuit to allow precise tuning of the comparator threshold. A 50 kΩ potentiometer served as the upper and lower portions of the divider, configured so that its wiper produced the scaled version of the supply voltage. With the supply set to the design trip voltage of 8 V, the potentiometer was adjusted until the difference between the Zener reference node and the divider node approached zero, visualized by a Keithley 2110 Digital Multimeter (DMM).

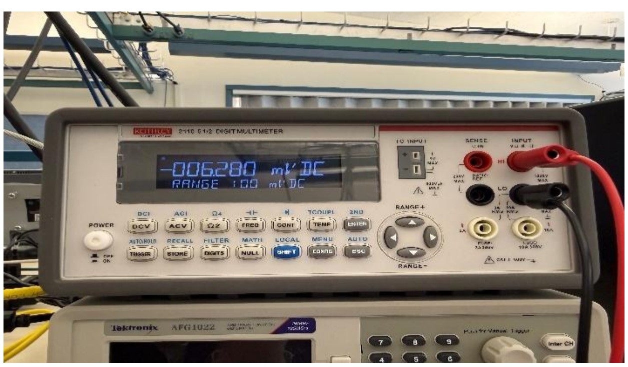

Fig. 2: DMM Reading of Voltage Difference

The final measured difference was −6.3 mV, indicating nearly perfect calibration.

C. *Procedure 3 – Comparator Switching*

In this stage, the TLC3702 comparator was added to the circuit following the schematic provided in the lab handout [1].

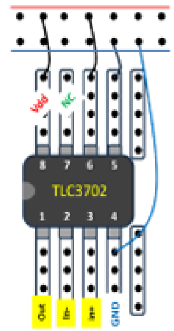

Fig. 3: Detailed Pinout of the TLC3702 Comparator [5]

One comparator channel was disabled, and the remaining channel was configured with the Zener reference connected to the negative input and the output of the divider connected to the positive input. Again, a DMM was used to record the output from the comparator.

TABLE II: Comparator Output from Procedure 3 Circuit

V_{DD}, V	**7**	**7.5**	**8.0**	**8.2**	**8.5**	**9.0**	**9.5**	**10**
V_C, mV	0.02	0.02	0.02	8002	8500	9000	9500	10000

As the supply voltage was gradually increased from 7 V upward, the comparator output was monitored. The output remained low until the supply reached approximately 8.02 V, at which point the comparator switched states, consistent with the predicted threshold based on the divider ratio and Zener reference voltage.

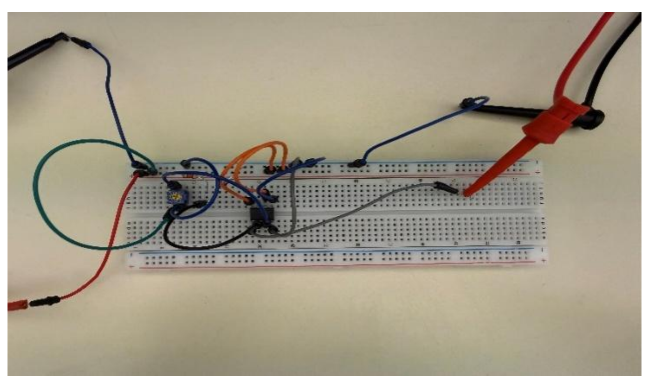

Fig. 4: Construction of Circuit from Procedure 3

A photograph of the corresponding breadboard setup was taken to document this stage.

D. Procedure 4– LED Activation

Following verification of the comparator's switching behavior, the LED indicator circuit was added to visibly represent the comparator output state.

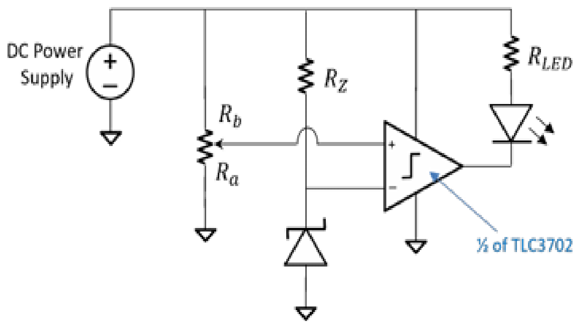

Fig. 5: Schematic of Low-Voltage Sensing Circuit with LED [1]

A current-limiting resistor R_{LED} was selected based on the LED's forward voltage characteristics from its datasheet [2], ensuring proper operation when driven by the comparator. When the supply voltage dropped below approximately 8.05 V, the comparator output transitioned low, allowing current through the LED and producing a clear red indication. When the supply exceeded the threshold, the LED turned off completely.

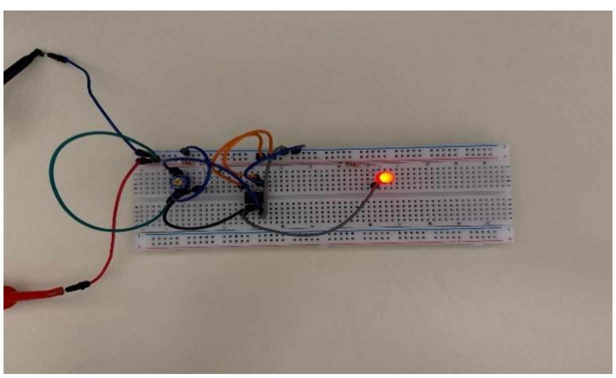

Fig. 6: Construction of Circuit from Procedure 4

A photograph showing the LED illuminated under low-supply conditions was taken to verify functionality.

III. Conclusion

The results from all parts of the experiment lined up well with what was expected, and each procedure helped confirm that the low-voltage detection circuit was working the way it should. Starting with the Zener reference, the measured voltage rose slightly from 3.40 V to 3.62 V as the supply increased. This small change is typical for a Zener under load and didn't interfere with its role as a stable comparison point. When the potentiometer divider was adjusted to match this reference, the difference at 8 V was only −6.3 mV, showing that the threshold could be set very close to the intended value.

With the reference and threshold established, the comparator switched at 8.02 V, which is close to the designed trip point once normal component tolerances and minor environmental factors are taken into account. The LED stage followed this behavior closely, turning on at about 8.05 V. This matched the comparator's output and confirmed that the LED current and forward-voltage characteristics were chosen appropriately.

Putting these results together, the Zener, divider, comparator, and LED behaved as a coherent system and produced a reliable undervoltage indication. The experiment also shows that a simple set of analog parts can still provide accurate and repeatable performance, even with the small variations that show up in real hardware. Overall, the circuit met its design goal, and the measured trip point near 8 V suggests it would work well in battery monitors or other low-voltage warning applications.

Acknowledgment

The author thanks Professor Ellis for his guidance throughout the course and during this laboratory assignment. Additional appreciation is extended to lab partners Keoni Galang and Bader Karam for their collaboration, support, and shared effort in completing the experimental work.

References

[1] N. Ellis, WEEK #7: A Low-Supply Voltage Sensor with an Optical Indicator, Cal Poly SLO, 2024

[2] Kingbright, WP7113ID Datasheet, Digikey.com, 2024. [Online]. Available: https://www.kingbrightusa.com/images/catalog/SPEC/WP7113ID.pdf K.

[3] Onsemi, 1N5221B – 1N5252B Zener Diodes Datasheet, Digikey.com 2016. [Online]. Available: https://www.onsemi.com/pdf/datasheet/1n5221b-d.pdf

[4] Texas Instruments, "TLC3702 DUAL MICROPOWER LinCMOS™ VOLTAGE COMPARATORS," Digikey.com, 2012. [Online]. Available: https://www.digikey.com/htmldatasheets/production/26981/0/0/1/tlc3702cdr.html?msockid=0aec75e0589e6043186a6103591d61bb

[5] N. Ellis, WEEK #6: Elementary Logic Circuits, Cal Poly SLO, 2024